AI Assisted Coding

Assignment – 3.2

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Question 1: Progressive Prompting for Calculator Design: Ask the AI to design a simple calculator

program by initially providing only the function name. Gradually enhance the prompt by

adding comments and usage examples.

Stage 1:

Code:

As_3.py

As_3.py > calculator

```python
# write a python function named calculator
def calculator(num1, num2, operation):
    if operation == 'add':
        return num1 + num2
    elif operation == 'subtract':
        return num1 - num2
    elif operation == 'multiply':
        return num1 * num2
    elif operation == 'divide':
        if num2 != 0:
            return num1 / num2
        else:
            return "Error: Division by zero"
    else:
        return "Error: Invalid operation"
# Example usage:
result = calculator(10, 5, 'add')
print(result)  # Output: 15
result = calculator(10, 5, 'subtract')
print(result)  # Output: 5
```

Stage 2:

Code:

```
Click to add a breakpoint
# write a python function named calculator
# the function should work as a simple calculator
# # it should take two numbers and an operator
# the operator can be +, -, *, /
def calculator(num1, num2, operator):
    if operator == '+':
        return num1 + num2
    elif operator == '-':
        return num1 - num2
    elif operator == '*':
        return num1 * num2
    elif operator == '/':
        if num2 != 0:
            return num1 / num2
        else:
            return "Error: Division by zero"
    else:
        return "Error: Invalid operator"
# Example usage:
result = calculator(10, 5, '+')
print(result)  # Output: 15
result = calculator(10, 5, '-')
print(result)  # Output: 5
result = calculator(10, 5, '*')
print(result)  # Output: 50
result = calculator(10, 5, '/')
print(result)  # Output: 2.0
result = calculator(10, 0, '/')
print(result)  # Output: Error: Division by zero
result = calculator(10, 5, '%')
print(result)  # Output: Error: Invalid operator
```

Output:

```
15
5
50
2.0
Error: Division by zero
Error: Invalid operator
```

Stage 3:

Code:

```python
# write a python function named calculator
# the function should work as a simple calculator
# # it should take two numbers and an operator
# the operator can be +, -, *, /
# it should return the result of the operation
# Example usage:
# calculator(10, 5, '+') should return 15
# calculator(10, 5, '-') should return 5
# calculator(10, 5, '*') should return 50
# calculator(10, 5, '/') should return 2.0
def calculator(num1, num2, operator):
    if operator == '+':
        return num1 + num2
    elif operator == '-':
        return num1 - num2
    elif operator == '*':
        return num1 * num2
    elif operator == '/':
        if num2 != 0:
            return num1 / num2
        else:
            return "Error: Division by zero"
    else:
        return "Error: Invalid operator"
# Example usage:
result = calculator(10, 5, '+')
print(result)  # Output: 15
result = calculator(10, 5, '-')
print(result)  # Output: 5
result = calculator(10, 5, '*')
print(result)  # Output: 50
result = calculator(10, 5, '/')
print(result)  # Output: 2.0
result = calculator(10, 0, '/')
print(result)  # Output: Error: Division by zero
result = calculator(10, 5, '%')
print(result)  # Output: Error: Invalid operator
```

Output:

```
15
5
50
2.0
```

Final Observation:

At first, when only the function name was given, the AI generated a very basic and incomplete calculator function with little or no logic. After adding comments, the AI started including parameters and arithmetic operations. When usage examples were finally added, the AI produced a complete and well-structured calculator program with proper conditions and error handling. This clearly shows that progressive prompting improves both the logic and structure of the generated code.

Question 2: Task Description-2

• Refining Prompts for Sorting Logic: Start with a vague prompt for sorting student marks, then refine it to clearly specify sorting order and constraints.

Expected Output-2

• AI-generated sorting function evolves from ambiguous logic to an accurate and efficient implementation.

Stage 1:

Code :

```python
# write a python program to sort student marks
def sort_student_marks(marks):
    return sorted(marks)
# Example usage:
student_marks = [88, 92, 79, 85, 95]
sorted_marks = sort_student_marks(student_marks)
print(sorted_marks)  # Output: [79, 85, 88, 92, 95]
```

Output :

```
3_2/Ass coding/As_3.py
[79, 85, 88, 92, 95]
```

Stage 2:

Code :

```python
# write a python functions to sort student marks in ascending order.
# the function should take a list of marks as input and return the sorted list.
def sort_student_marks(marks):
    return sorted(marks)
# Example usage:
student_marks = [88, 92, 79, 85, 95]
sorted_marks = sort_student_marks(student_marks)
print(sorted_marks)  # Output: [79, 85, 88, 92, 95]
```

Out put :

```
3_2/Ass coding/As_3.py
[79, 85, 88, 92, 95]
```

Stage 3:

Code:

```python

# write a python functions to sort student marks.
# implement a simple bubble sort algorithm to sort the marks in descending order.
def sort_student_marks_descending(marks):
    n = len(marks)
    for i in range(n):
        for j in range(0, n-i-1):
            if marks[j] < marks[j+1]:
                marks[j], marks[j+1] = marks[j+1], marks[j]
    return marks
# Example usage:
student_marks = [88, 92, 79, 85, 95]
sorted_marks = sort_student_marks_descending(student_marks)
print(sorted_marks)  # Output: [95, 92, 88, 85, 79]
```

Final Observation:

With a vague prompt, the AI produced a simple sorting solution without clear direction or constraints. After refining the prompt to specify sorting order, the output became more accurate and meaningful. When clear constraints and examples were added, the AI

generated a more structured and efficient sorting function. This demonstrates that refining prompts helps the AI move from ambiguous logic to a correct and reliable implementation.

Question 3: Task Description-3

• Few-Shot Prompting for Prime Number Validation: Provide multiple input-output examples for a function that checks whether a number is prime. Observe how few-shot prompting improves correctness.

Expected Output-3

• Improved prime-checking function with better edge-case handling.

Stage 1:

Code:

```python


#Write a Python function to check whether a number is prime.
def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True
#Example usage:
result = is_prime(11)
print(result)  # Output: True
result = is_prime(4)
print(result)  # Output: False

```

Output:

```
True
False
```

Stage 2:

Code:

```python


#Write a Python function to check whether a number is prime.
def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True
#Example usage:
result = is_prime(11)
print(result)  # Output: True
result = is_prime(4)
print(result)  # Output: False

```

Output:

```
True
False
```

Stage 3:

Code:

```python

# write a python function to check whether a number is prime.
# examples:
# input : 2 output : prime
# input : 3 output : prime
# input : 4 output : not prime
# input : 5 output : prime
# input : 6 output : not prime
# input : 7 output : prime
# input : 8 output : not prime
def is_prime(num):
    if num <= 1:
        return "not prime"
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return "not prime"
    return "prime"
# Example usage:
result = is_prime(11)
print(result)  # Output: prime
result = is_prime(4)
print(result)  # Output: not prime
result = is_prime(2)
print(result)  # Output: prime
result = is_prime(9)
print(result)  # Output: not prime
result = is_prime(13)
print(result)  # Output: prime
result = is_prime(1)
print(result)  # Output: not prime
result = is_prime(0)
print(result)  # Output: not prime
result = is_prime(-5)
print(result)  # Output: not prime
```

Output:

```
prime
not prime
prime
not prime
prime
not prime
not prime
not prime
```

Final Observation:

In the initial prompt without examples, the AI generated a basic prime-checking function that could miss important edge cases. When one example was provided, the result improved slightly. After giving multiple input-output examples (few-shot prompting), the AI clearly handled cases like 0, 1, and negative numbers and produced a more accurate and robust prime-checking function. This shows that few-shot prompting improves correctness and edge-case handling.

Question 4: Task Description-4

• Prompt-Guided UI Design for Student Grading System: Create a user interface for a student grading system that calculates total marks, percentage, and grade based on user input.

Expected Output-4

• Well-structured UI code with accurate calculations and clear output display.

Stage 1:

Code:

```python
# create a user interface for a student grading system
def student_grading_system():
    def calculate_grade(marks):
        if marks >= 90:
            return 'A'
        elif marks >= 80:
            return 'B'
        elif marks >= 70:
            return 'C'
        elif marks >= 60:
            return 'D'
        else:
            return 'F'

    print("Welcome to the Student Grading System")
    name = input("Enter student name: ")
    try:
        marks = float(input("Enter student marks (0-100): "))
        if 0 <= marks <= 100:
            grade = calculate_grade(marks)
            print(f"Student Name: {name}")
            print(f"Marks: {marks}")
            print(f"Grade: {grade}")
        else:
            print("Error: Marks should be between 0 and 100.")
    except ValueError:
        print("Error: Invalid input. Please enter numeric marks.")
# Example usage:
student_grading_system()
```

Stage 2:

Code:

```python
# create a simple user interface for a student grading system using python
# the user should enter marks and the program should calculate total and percentage
def student_grading_system():
    print("Welcome to the Student Grading System")
    name = input("Enter student name: ")
    try:
        marks_obtained = float(input("Enter marks obtained: "))
        total_marks = float(input("Enter total marks: "))
        if 0 <= marks_obtained <= total_marks:
            percentage = (marks_obtained / total_marks) * 100
            print(f"Student Name: {name}")
            print(f"Marks Obtained: {marks_obtained}")
            print(f"Total Marks: {total_marks}")
            print(f"Percentage: {percentage:.2f}%")
        else:
            print("Error: Marks obtained should be between 0 and total marks.")
    except ValueError:
        print("Error: Invalid input. Please enter numeric values for marks.")
# Example usage:
student_grading_system()
```

Stage 3:

Code:

```python
# create a python tkinter user interface for a student grading system

# requirements:
# ui should accepet student name and marks of 5 subjects
# each subject marks should be out of 100
# on submit it should calculate total marks,percentage and grade

# grade rules:
# 90-100 : A
# 80-89  : B
# 70-79  : C
# 60-69  : D
# below 60 : F
```

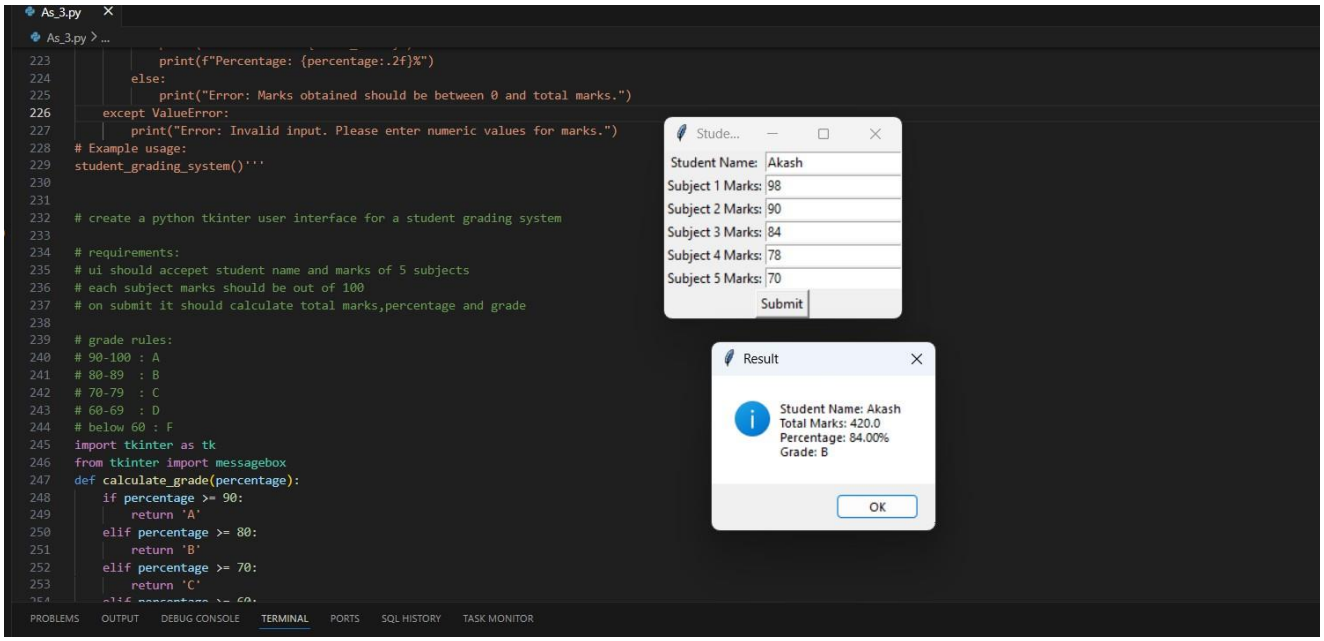
```python
import tkinter as tk
from tkinter import messagebox
def calculate_grade(percentage):
    if percentage >= 90:
        return 'A'
    elif percentage >= 80:
        return 'B'
    elif percentage >= 70:
        return 'C'
    elif percentage >= 60:
        return 'D'
    else:
        return 'F'
def submit():
    try:
        name = entry_name.get()
        marks = [float(entry.get()) for entry in entries_marks]
        if any(mark < 0 or mark > 100 for mark in marks):
            messagebox.showerror("Error", "Marks should be between 0 and 100.")
            return
        total_marks = sum(marks)
        percentage = (total_marks / 500) * 100
        grade = calculate_grade(percentage)
        result = f"Student Name: {name}\nTotal Marks: {total_marks}\nPercentage: {percentage:.2f}%\nGrade: {grade}"
        messagebox.showinfo("Result", result)
    except ValueError:
        messagebox.showerror("Error", "Invalid input. Please enter numeric values for marks.")
root = tk.Tk()
root.title("Student Grading System")
tk.Label(root, text="Student Name:").grid(row=0, column=0)
entry_name = tk.Entry(root)
entry_name.grid(row=0, column=1)
entries_marks = []
for i in range(5):
    tk.Label(root, text=f"Subject {i+1} Marks:").grid(row=i+1, column=0)
    entry = tk.Entry(root)
    entry.grid(row=i+1, column=1)
    entries_marks.append(entry)
tk.Button(root, text="Submit", command=submit).grid(row=6, columnspan=2)
root.mainloop()
```

Output:

Task_01.py Task_03.py

Task_03.py > ...

```python
btn_calculate.grid(row=6, columnspan=2)
# Run the application
root.mainloop()
```



Final Observation:

With a vague UI prompt, the AI produced only a simple or unclear interface idea. As the prompt was refined to include calculation requirements, the UI output became more meaningful. When full instructions were given (inputs, calculations, grade rules, and display), the AI generated a well-structured user interface with correct total, percentage, and grade calculation along with clear result display. This shows that prompt guidance greatly improves UI structure and usability.

Question 5: Task Description-5

• Analyzing Prompt Specificity in Unit Conversion Functions: Improving a Unit Conversion Function (Kilometers to Miles and Miles to Kilometers) Using Clear Instructions.

Expected Output-5

• Analysis of code quality and accuracy differences across multiple prompt variations.

Stage 1:

Code:

```python

# write a python function to convert units
def convert_units(value, from_unit, to_unit):
    conversion_factors = {
        'meters_to_kilometers': 0.001,
        'kilometers_to_meters': 1000,
        'grams_to_kilograms': 0.001,
        'kilograms_to_grams': 1000,
        'celsius_to_fahrenheit': lambda c: (c * 9/5) + 32,
        'fahrenheit_to_celsius': lambda f: (f - 32) * 5/9
    }
    key = f"{from_unit}_to_{to_unit}"
    if key in conversion_factors:
        factor = conversion_factors[key]
        if callable(factor):
            return factor(value)
        else:
            return value * factor
    else:
        return "Error: Invalid conversion"
# Example usage:
result = convert_units(1000, 'meters', 'kilometers')
print(result)  # Output: 1.0
result = convert_units(5, 'kilograms', 'grams')
print(result)  # Output: 5000
result = convert_units(100, 'celsius', 'fahrenheit')
print(result)  # Output: 212.0
result = convert_units(212, 'fahrenheit', 'celsius')
print(result)  # Output: 100.0
result = convert_units(10, 'meters', 'grams')
print(result)  # Output: Error: Invalid conversion
```

Output:

```
1.0
5000
212.0
100.0
Error: Invalid conversion
```

Stage 2:

Code:

```python
# write a python function to convert kilometers to miles
def kilometers_to_miles(km):
    miles = km * 0.621371
    return miles
# Example usage:
result = kilometers_to_miles(10)
print(result)  # Output: 6.21371
result = kilometers_to_miles(5)
print(result)  # Output: 3.106855
```

Stage 3:

Code:

```python
# create a python function for unit conversion
# the function should convert kilometers to miles and miles to kilometers
def convert_distance(value, from_unit, to_unit):
    if from_unit == 'kilometers' and to_unit == 'miles':
        return value * 0.621371
    elif from_unit == 'miles' and to_unit == 'kilometers':
        return value / 0.621371
    else:
        return "Error: Invalid conversion"
# Example usage:
result = convert_distance(10, 'kilometers', 'miles')
print(result)  # Output: 6.21371
result = convert_distance(6.21371, 'miles', 'kilometers')
print(result)  # Output: 10.0
result = convert_distance(5, 'kilometers', 'grams')
print(result)  # Output: Error: Invalid conversion
```

Final Observation:

When a vague prompt was used, the AI generated unclear or very general conversion code. After specifying the type of conversion, the AI produced a basic one-way converter. When detailed instructions, formulas, and validation rules were added, the AI generated an accurate, well-structured, and reusable unit conversion function. This proves that higher prompt specificity leads to better code quality, accuracy, and reliability.